

Student Name: Chamikara W.A.Y
Student ID: 28341

Question 1

a)

While AI, ML, and DL are interrelated concepts, they are still not quite the same things. First of all, AI is an umbrella term for technologies whose main goal is to build machines able to do work usually done by people. ML refers to learning algorithms used by computers without any prior programming. Finally, DL refers to learning algorithms utilizing artificial neural networks with more than one layer of processing nodes.

Interconnection Diagram:

Artificial Intelligence (AI)
└─ Machine Learning (ML)
 └─ Deep Learning (DL)

b)

Reinforcement Learning (RL) is a type of machine learning where an agent learns by interacting with an environment. The agent performs actions and receives rewards or penalties. Over time, the agent learns the best strategy to maximize rewards.

Applications:

- Robotics
- Self-driving cars
- Game playing systems
- Recommendation systems

Example: A self-driving car learns how to safely drive by receiving positive rewards for following traffic rules and penalties for unsafe actions.

c)

1. Generative AI and Large Language Models (LLMs): Systems such as ChatGPT can generate human-like text and assist in coding, education, and customer support.

2. AI in Healthcare: Deep learning models help detect diseases such as cancer from medical images with high accuracy.

3. Autonomous Systems: AI-powered robots and autonomous vehicles improve transportation, logistics, and industrial automation.

Question 2

1.

a) The way through which knowledge is represented for use by an AI machine refers to knowledge representation. Knowledge representation is crucial because through it AI agents can make decisions and perform logical reasoning.

b) Two techniques used in medical diagnosis:

- Rule-based representation: Uses IF-THEN rules.
- Semantic networks: Represents relationships between diseases and symptoms.

2.

a) A knowledge-based agent is an AI system that uses stored knowledge and reasoning techniques to make decisions. Unlike simple reflex agents, it can reason about situations using stored facts.

b) Components:

- Knowledge Base: Stores facts and rules.
- Inference Engine: Applies logical reasoning.
- Knowledge Acquisition Module: Updates knowledge.
- User Interface: Communicates with users.

3.

a) Logical reasoning is the process of drawing conclusions from known facts and rules. It is essential in knowledge-based systems for diagnosis and decision-making.

b) Logical operators such as AND, OR, and NOT are used to form logical statements. Inference rules such as Modus Ponens help derive conclusions from premises.

4.

a) DeMorgan's Theorem:

$$\neg(A \wedge B) = (\neg A \vee \neg B)$$

$$\neg(A \vee B) = (\neg A \wedge \neg B)$$

It simplifies logical expressions and is important in logical reasoning.

b) Bayes' Theorem calculates conditional probability:

$$P(A|B) = [P(B|A) \times P(A)] / P(B)$$

In medical diagnosis, it helps estimate the probability of a disease given observed symptoms.

5.

a) Probability measures the likelihood of an event occurring. In medical diagnosis, it helps estimate the likelihood of diseases.

b) Probability theory helps doctors and AI systems make decisions under uncertainty by estimating risks and selecting the most likely diagnosis.

Question 3

Tree 1:-

BFS order

Queue-based level-order traversal starting from A:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I \rightarrow J \rightarrow K$

(From left to right level by level)

A, B, C, D, E, F, G, H, I, J, K

DFS order

DFS using usual recursive pre-order (visit node, then children left to right):

$A \rightarrow B \rightarrow D \rightarrow H \rightarrow I \rightarrow E \rightarrow C \rightarrow F \rightarrow J \rightarrow G \rightarrow K$

Answer for DFS Tree 1:

A, B, D, H, I, E, C, F, J, G, K

UCS order

A, B, D, C, E, I, F, J, G, H, K

Tree 2 BFS order

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O

DFS order (pre-order left-to-right)

A, B, D, H, I, E, J, K, C, F, L, M, G, N, O

UCS order

Order: A, B, C, E, D, G, J, F, O, I, N, H, K, L, M

Tree 3:-

BFS order

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I \rightarrow J$

BFS Tree 3: A, B, C, D, E, F, G, H, I, J

DFS order (pre-order left-to-right)

$A \rightarrow B \rightarrow D \rightarrow H \rightarrow I \rightarrow E \rightarrow C \rightarrow F \rightarrow J \rightarrow G$

A, B, D, H, I, E, C, F, J, G

UCS Tree A, B, E, D, C, H, I, F, G, J

Tree 4:-

BFS order

A, B, C, D, E, F, G, H, I

DFS order

A, B, D, F, H, I, G, E, C

UCS Tree A, B, E, C, D, G, F, H, I

Tree 5:

BFS order

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I \rightarrow J$

DFS order

$A \rightarrow B \rightarrow D \rightarrow G \rightarrow H \rightarrow E \rightarrow I \rightarrow J \rightarrow C \rightarrow F$

UCS Tree A, C, B, D, F, G, E, H, I, J

Breadth-First Search (BFS) explores level by level. Depth-First Search (DFS) explores deeply before backtracking. Uniform-Cost Search (UCS) expands the lowest-cost node first.

Question 4

a)

i) $(A \wedge B) \Rightarrow (A \vee B)$

To determine whether the statement is valid, satisfiable, or unsatisfiable, we construct a truth table.

A	B	$A \wedge B$	$A \vee B$	$(A \wedge B) \Rightarrow (A \vee B)$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

The statement is true for all possible truth assignments.

Therefore, $(A \wedge B) \Rightarrow (A \vee B)$ is a Valid statement (Tautology).

ii) $(A \wedge B) \wedge (\neg A \wedge \neg B)$

This expression requires A and $\neg A$ to be true simultaneously and B and $\neg B$ to be true simultaneously.

$[(A \wedge B) \wedge (\neg A \wedge \neg B)] = [(A \wedge \neg A) \wedge (B \wedge \neg B)]$

Since $(A \wedge \neg A)$ is always false, the entire expression is always false.

Therefore, the statement is Unsatisfiable (Contradiction).

iii) $(A \Rightarrow B) \Rightarrow (B \Rightarrow A)$

Constructing the truth table:

A	B	$A \Rightarrow B$	$B \Rightarrow A$	$(A \Rightarrow B) \Rightarrow (B \Rightarrow A)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

The statement is true for some truth assignments and false for others.

Therefore, the statement is Satisfiable but not Valid.

b) **Given Statements**

1. $\neg(\neg P \vee Q)$

2. $\neg(S \vee \neg T)$
3. $\neg T \wedge \neg Q$
4. $(\neg U \vee (T \vee (S \wedge \neg P)))$

Step 1: Convert to Clause Form (CNF)

Statement 1

$$\neg(\neg P \vee Q)$$

Applying DeMorgan's Law:

$$[P \wedge \neg Q]$$

Clauses:

1. P
2. $\neg Q$

Statement 2

$$\neg(S \vee \neg T)$$

Applying DeMorgan's Law:

$$[\neg S \wedge T]$$

Clauses:

3. $\neg S$
4. T

Statement 3

$$[\neg T \wedge \neg Q]$$

Clauses:

5. $\neg T$
6. $\neg Q$

Statement 4

$$[(\neg U \vee (T \vee (S \wedge \neg P)))]$$

Distributing:

$$[(\neg U \vee T \vee S)]$$

$$[(\neg U \vee T \vee \neg P)]$$

Clauses:

$$7. \neg U \vee T \vee S$$

$$8. \neg U \vee T \vee \neg P$$

Step 2: Apply Resolution

Resolve Clause (8) with Clause (1):

$$(\neg U \vee T \vee \neg P)$$

and

$$[P]$$

Result:

$$9. \neg U \vee T$$

Resolve Clause (9) with Clause (5):

$$[(\neg U \vee T)]$$

and

$$[\neg T]$$

Result:

$$10. \neg U$$

Final Answers

a)

i. $(A \wedge B) \Rightarrow (A \vee B) \rightarrow$ **Valid (Tautology)**

ii. $(A \wedge B) \wedge (\neg A \wedge \neg B) \rightarrow$ **Unsatisfiable (Contradiction)**

iii. $(A \Rightarrow B) \Rightarrow (B \Rightarrow A) \rightarrow$ **Satisfiable but not Valid**

b)

Using Resolution Refutation, the conclusion obtained is:

$\boxed{\neg U}$

Hence, $\neg U$ is **proved**.